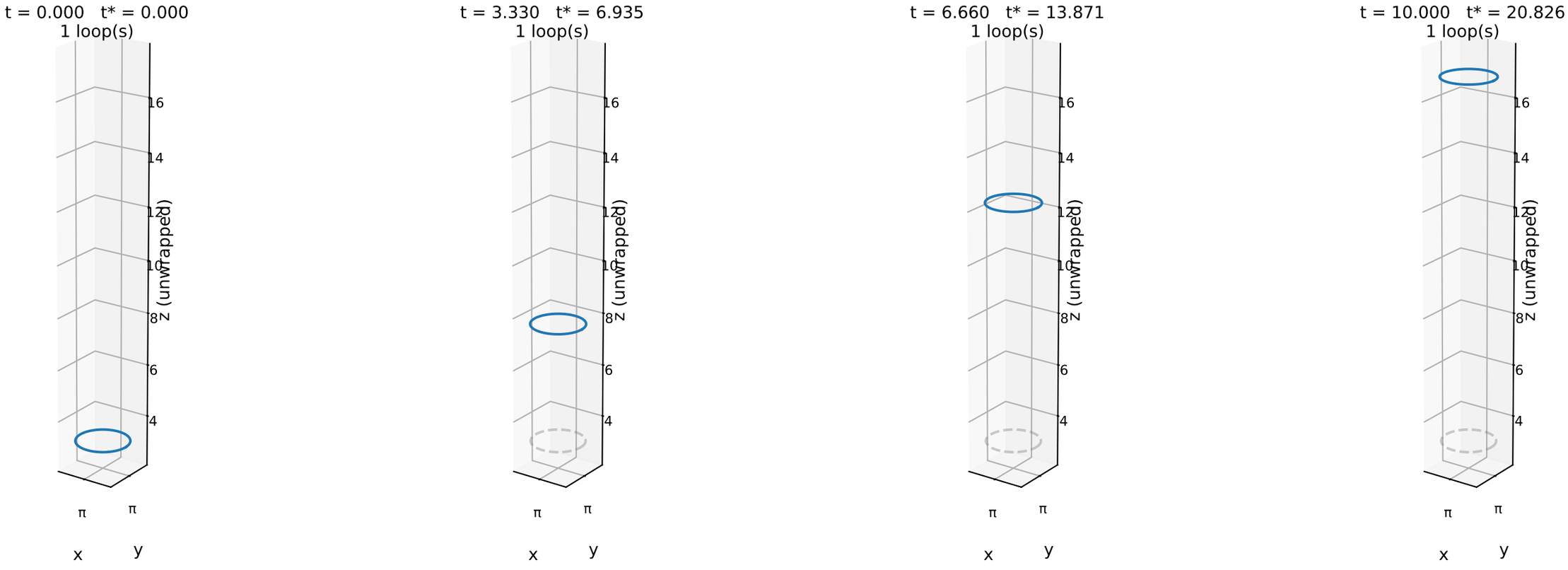single_ring · fixed laboratory frame · equal physical coordinate scales